

GANESH GUPTA

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Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Design enterprise RAG architecture (chunking, embeddings, hybrid retrieval, reranking) grounding multiple LLMs (GPT, Claude, Llama, Gemini) in enterprise knowledge sources.
- Build agentic AI with tool/function calling (LangGraph, CrewAI, AutoGen, OpenAI Agents SDK), deployed on Azure and AWS with production CI/CD.
- Own AI model evaluation, prompt versioning, and model lifecycle management, enforcing guardrails and cost optimization across releases.
- Build REST APIs (Python, FastAPI) and full-stack platform features, partnering with product and business stakeholders end-to-end.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Developed multi-modal deep learning models (PyTorch) for medical imaging analysis, and ran hands-on model fine-tuning on multi-GPU clusters with compression and quantization.
- Built PySpark data processing jobs and custom evaluators for RAG/agent workflows, tracking drift and regression across model versions before production.
- Improved semantic retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Mentored engineers through code reviews and design write-ups, driving consistent engineering standards.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with MCP-compatible tool-use compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP server for enterprise data queries.

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, SQL, TypeScript, Java, C++

GenAI & Agentic AI: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK, GPT, Claude, Llama, Gemini, Tool/Function Calling

RAG & ML: Enterprise RAG, Chunking, Embeddings, Hybrid Retrieval, PyTorch, Model Fine-Tuning, PGVector/PostgreSQL

Cloud & DevOps: Azure, AWS, FastAPI, Docker, Kubernetes, CI/CD, Monitoring

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer